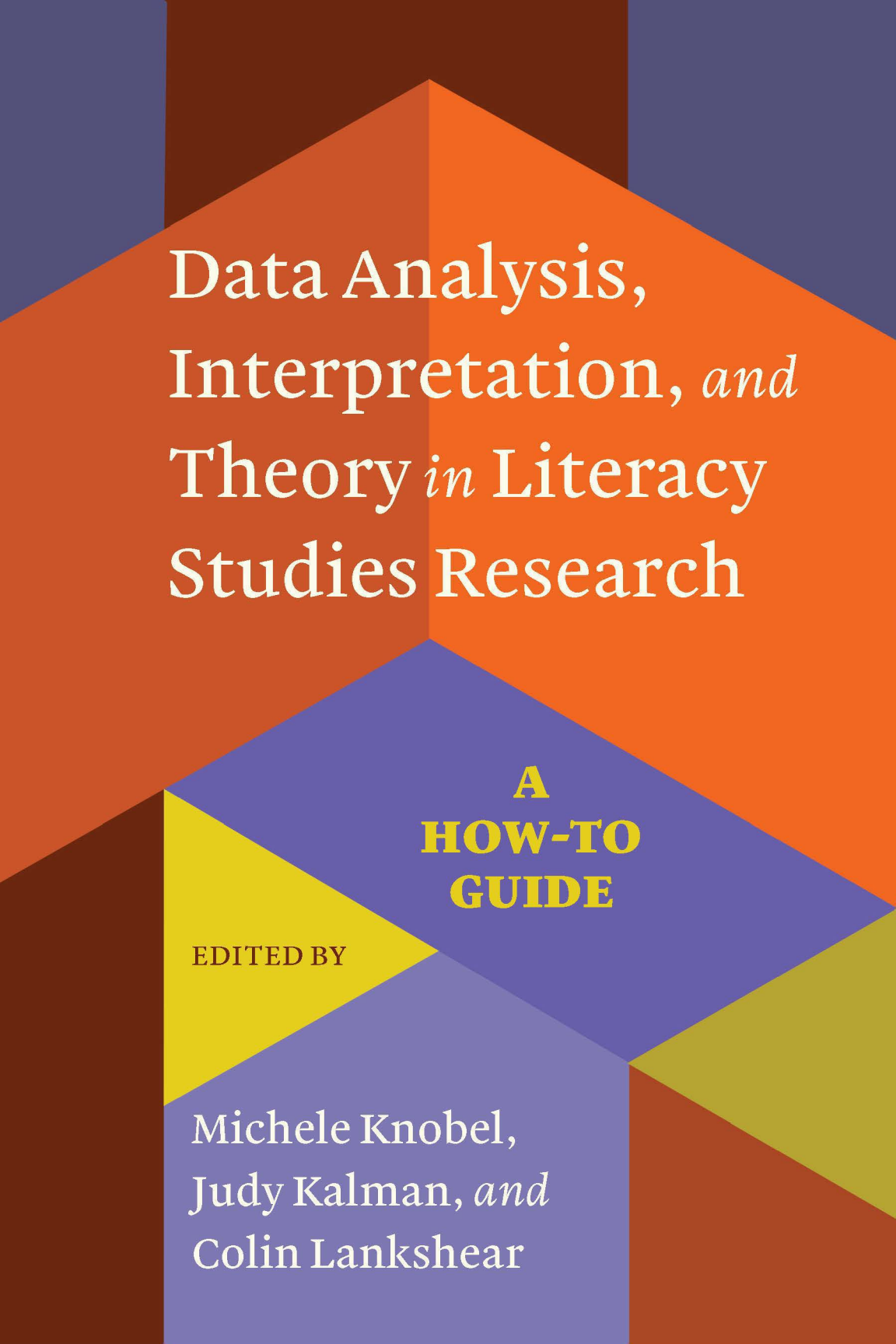




Data Analysis, Interpretation, *and* Theory *in* Literacy Studies Research

A HOW-TO GUIDE

EDITED BY

Michele Knobel,
Judy Kalman, *and*
Colin Lankshear

DATA ANALYSIS, INTERPRETATION, AND
THEORY IN LITERACY STUDIES RESEARCH

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and Colin Lankshear*



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CHAPTER 10

Following Texts, Bodies, and Objects in Everyday Practice: Thinking With Actor Network Theory

Jason F. Lovvorn and Kevin M. Leander

EDUCATIONAL THEORIES, LIKE many social theories, largely remain people-centered. At the same time, they often inherit and maintain structuralism's penchant for particular forms of stability, sorting experience and creating order out of disorder while prioritizing humanistic patterns of interpretation. In part, such tendencies derive from the inherently human associations, no doubt created and sustained by humans, that surround educational practice. Teachers teach. Students learn. Administrators administer. In this disposition, agents of practice and agents of change are grounded in psycho-social-cultural perspectives wherein human intentions take center stage and material objects exist largely as tools (e.g., Cole & Engestrom, 1993; Engestrom, 1999). In such models, shaped by what Latour (2005) calls a "strong 'socio-logy,'" human subjectivity reigns supreme, and people own the verbs put into play.

More recent scholarly efforts, however, challenge human primacy, going so far as to suggest an active, equal footing for nonhuman materials. This "posthuman" turn (Barad, 2003; Braidotti, 2013) posits that nonhumans share ability and should share credit for work in the world, including the knowledge work of educational practice. In this context, a posthuman "New Materialism" (Barrett & Bolt, 2013; Coole & Frost, 2010; Dolphijn & van der Tuin, 2012) has emerged across multiple disciplines, focusing attention on the ways in which *things* hold sway. In this vein, "thing theory" (Brown, 2001) has helped

scholars reorganize the ways in which we understand agency and accountability (Bennett, 2010; Brown, 2004; Lamb, 2011). *Vibrancy* and *vitality* are descriptors ascribed to nonhuman objects in these new accounts where things possess force (Bennett, 2010). Here, objects share verbs traditionally wielded by humans. Spaces mark. Bodies influence. Things shape.

A specific strand of materially-oriented work drives this chapter. As part of the shift toward an “object-oriented” (Harman, 2009) philosophy/ontology in social research, Actor Network Theory (hereafter abbreviated as ANT) originated with ethnographic efforts to understand how scientific knowledge arises across identifiable contexts (Latour, 1987, 1993; Latour & Woolgar, 1986). This work in the sociology of science considers how various agents, both human and nonhuman, are distributed in complex networks, making possible certain forms of knowing. For example, in thinking about Einstein’s famous equation $E = mc^2$, an ANT account would consider Einstein’s genius as merely one factor contributing to this breakthrough. Besides the intellectual tradition preceding Einstein, ANT might also describe nonhuman actors, such as the city and university libraries in Bern, a drawer in the Bern patent office where Einstein kept his notes, and even the equals sign itself, a symbol that did not come about until the 16th century. Bruno Latour (1988), a key originator of ANT, in fact, examines Einstein’s theory of relativity with an eye toward what he calls the “sociology of delegation,” wherein the origins of the treatise stretch widely across a range of objects and humans. In this regard, Latour lists among the important “actants” (his term for *any* element that *acts*) the following: rods, a train, Trafalgar Square, a raven, fixed stars, lightning, a marble table, a railway carriage, stones, and a note emitted by an organ pipe.

ANT therefore requires of the researcher an ability to level distinctions between people and things. This flattening—whether the metaphor is delegation (Latour, 1988), distribution (Bennett, 2010), or symmetry (Callon, 1986; Spinuzzi, 2015)—insists that objects be granted agency equal to that which humans enjoy in people-centered descriptions of activity. As Spinuzzi (2015) succinctly defines it, such symmetry is “the principle that human and nonhuman actants are treated alike when considering how controversies are settled” (p. 23). Three important implications arise from this line. First, as already noted, humans and nonhumans share agency. Second, the interconnections and play of these varied “actants” shape how knowledge settles into view. Third, such settlements are not predetermined; they involve shifts and struggles as actants play off one another and therefore require careful tracking at the ground level. Combined, such ideas push us from ANT as social theory toward a more layered understanding of this domain as both theory and methodology.

As a tool in the employ of the literacy researcher, ANT defies easy categorization. Part theory, part method, it signals an attitude toward activity at the same time it recommends descriptive techniques. Its originators recognize as much. Latour (1999a), for instance, notes, “For us, ANT was simply another way of being faithful to the insights of ethnomethodology: actors know what they do and we have to learn from them not

only what they do, but how and why they do it” (p. 19). Thus, even though “theory” is foregrounded in ANT’s name, just as much emphasis rests on *how* researchers might trace significant elements in activity. More recent scholarship echoes this sentiment. For example, Nespore (2012) points out that ANT is both “a set of assumptions and conceptual tools for studying them” (p. 2), and Hamilton (2012) positions this work as a “conversation between theory and data” (p. 41). Fenwick and Edwards (2012) stretch these insights even further, noting, “ANT is not ‘applied’ like a theoretical technology, but is more like a sensibility, a way to sense and draw (nearer to) a phenomenon” (p. ix). Invariably grounded in case studies, ANT forces a simultaneous consideration of theory and data, often asking investigators to consider novel ways to categorize and understand data. Such was the case in our prior examination of literacy practices in and out of school, the centerpiece for the methodological discussion that follows.

As a kind of sensibility, our attraction to ANT in this work was largely based on wanting to try and understand something that felt very abstract (space-time relations and literacy) in the more material and embodied experiences of using literacy. At the time of this research (approximately 2004 to 2006), it was unclear how conceptions of in-school, out-of-school, or other setting-based separations provide powerful distinctions for understanding differences among practices of literacy learning, teaching, and thinking. Largely, we think these puzzles are still with us, although we have made some progress and like to think of this study as one contribution. Literacy theory, we want to point out, always engages some theory or concept of space-time, and, as Brandt and Clinton (2002) have argued, we must acknowledge the “heavy hand literacy has had in building networks across time and space” (p. 347). Texts, a mainstay of literacy research, coordinate and structure space-time within a given activity and space-times across diverse activities. For researchers interested in understanding this reach, an important theoretical and practical task is to interpret the configurations, circulation, and translations of texts in relation to other actors in activity. ANT offers both theory and method for doing so.

Moreover, practices of reading and writing are invariably interwoven with other forms of activity. Literacy doesn’t only exist in some set of “events” that are separate from the events of everyday living and moving and doing all manner of things—this idea was becoming a mainstay finding in the New Literacy Studies. In moving social practice theories of literacy (Gee, 1990; Street, 1984) forward, we argue, researchers need to develop a view of the social that includes an understanding of how a wide range of actors (e.g., humans, texts, technologies, and objects) are brought into relationship with one another, with a careful eye on how these relationships have unique space-time qualities. This movement will also extend emerging conversations about how literacy and texts participate in social-material configurations (Fenwick & Edwards, 2012) and encourage researchers to broaden the scope of what is meant by *literacy*.

Actor Network Theory in Action: Summary of “Literacy Networks”

“Literacy Networks” (Leander & Lovvorn, 2006) is a particular kind of research article that differs somewhat from typical article norms, including posing a research question or questions, selecting a theoretical lens for the question(s), and then applying this lens for the analysis of the data. Rather, the study was more of an investigation of theory and data together, in an attempt to create new ways of understanding these data, with the hopes that these new ways might have some traction in the field beyond. In that sense, then, the connections among theory, method, and data were messy in this study, and we try to offer an unsanitized view of this messiness in our account as we move along.

The questions guiding the study, although they are more implied than stated outright in publication, were the following:

1. How might the study of networks, drawing on ANT, help us to think about literacy practices and their relations to space-time? Can we conceptualize something like a “literacy network?”
2. How might the analysis of literacy networks (if we can pull that off) help us to understand the engagement, agency, and identity of one youth across his different practices with literacy?

Looking back on these questions and directions now, more than 13 years since the study’s publication, they seem ambitious and complicated, which we think also “locates” this study as a kind of exploration across theory, method, and data—as an attempt to open up some new ways of thinking—without reaching something like “tested” arguments for seeing the data one way over all others. In that sense, work in the study was sometimes more divergent than convergent and drew on our sense of imagination as well as our analytic capacities.

Findings reported in our “Literacy Networks” article (Leander & Lovvorn, 2006) resulted from a case study that was part of a larger study called “Synchrony,” itself structured as a set of seven case studies that followed youth participants across school and online contexts. Synchrony was an effort to understand how the use of (then “new”) information and communication technologies, such as instant messaging, chat, e-mail, searching the Internet, building websites, and online gaming was related to youth literacy practices, to their social networks, and to their identities in offline contexts, including but not limited to school. Synchrony was not only fascinating but also difficult and very messy to write about at any scale, and one way to deal with this messiness was to select individual cases, such as the case of “Brian” presented in “Literacy Networks.” Brian was a 13-year-old youth during the time of the research and lived with his brother and parents outside a small town (population 6,000) in the mid-South of the United States. Of all the cases in Synchrony, Brian’s was the most robust, in part, because he was the most willing to engage in the research and stayed with it longer than the other participants.

We collected data to develop a case study of Brian's activities with literacy across school and home contexts through standard ethnographic methods: participant observations, document collection, interviews, small surveys of his classmates, and field notes. To record Brian's online activity, which, in his case, was nearly entirely from home, we used software for the digital capture of online interaction, and we videotaped his online interaction using a digital video camera typically placed over Brian's shoulder. Brian was observed at least once per week in his English and history classes from February to June 2003, and his course work from these classes was collected during these visits. Biweekly home observations of online literacy practices, including computer gaming, also began in February and continued through August 2003, with less frequent follow-up visits extending for several months.

Brian had little competition from family members for access to a recent model PC with a cable modem. He spent, on average, two to three hours per day in online activity, which primarily involved playing massively multiplayer online games (MMOGs). During the summer months, Brian's online activity often increased to five or more hours per day as he began to play *Star Wars Galaxies: An Empire Divided* (Lucas Arts & Sony Entertainment, 2003), an MMOG based loosely on the *Star Wars* milieu.

Analysis in Practice: Bringing Constructs From Theory and Selections From Data Into Dialogue

Before we launch into the nitty-gritty of our analysis-in-practice, it seems important to note a few things that had an important influence on our analysis but may not be explicit in reading the original paper alone. "Literacy Networks" (Leander & Lovvorn, 2006) is a study out of a larger study (Synchrony) that was exploring literacy practices situated in and out of school, on one hand, and online and offline, on the other. So a first observation about our larger method is that we were engaged in a comparative method within our ethnographic work—looking for differences and similarities across these boundaries (in/out, online/offline). We had chosen schools where there was at least some engagement online, so by design, we had two sets of comparisons going that did not neatly map onto one another.

In addition to this comparative method, and inspiring it, was a concept from spatial theory that we were continually thinking about: *How can we think beyond the notion of activities simply being "situated" in space and time? Or how does literacy relate to space-time?* The idea of literacy as "situated" was a major idea in Literacy Studies when we wrote the article (and still is). The idea of situatedness, in fact, was itself pushing back on something more traditional. The New Literacy Studies had pushed away from abstract or "autonomous" theories of literacy and subscribed instead to "ideological" theories (Street, 1984). Part of the meaning of thinking about literacy as "ideological," in the New Literacy Studies, was to consider how literacy was "situated" in the everyday lives of people. This

situatedness often was taken up through a focus on local practices, such as is the bread and butter of much of anthropology. Through a nascent critique of “situatedness,” we were inclined in the larger study to keep thinking about space and time relations, including the movements of people, things, and texts. So, space, time, and movements were key mindsets or forms of “framing” with which we continually looked at our data. In our analysis, we wanted to mess with “situatedness,” and especially with container-like views that simply saw texts or practices with literacy as being located somewhere.

In service of our larger question about how literacy may be actively involved in producing space-time relations (and not just situated in “contexts”), and in service of the more focused questions of the study, we developed for our analysis five related constructs. These constructs were informed by our reading of ANT, but our reading of the theory was also related to our questions and to our data. These constructs need emphasizing because it would be inaccurate to say that the following five qualities somehow appeared wholly formed and ready to use in the theory, as a kind of “model” or template for analysis. It would also be inaccurate to say that we created a theoretical model outside of the context of having data and thinking about these data—like a ready-made “toolkit” or “lens” to apply to the data. Rather, these five constructs were our attempt to put part of the theory to work for us—to clarify for ourselves how and when we would “see” ideas within the data that we had already collected but were still selecting cases and ideas from. We see data collection and selection, theory development, and data analysis as deeply intermingled in this work.

We developed five analytic dimensions from ANT during the course of the study (see Leander & Lovvorn, 2006, pp. 305–306):

1. Translation
2. Heterogeneity of space-time representations
3. Movements and positions of texts in circulation
4. Rhythms and speeds of circulation
5. Network continuity

Although we referred to these dimensions as a “framework” in the original article, we hesitate to do so here because a framework implies something more static and completely formed.

We mainly addressed textual translation, although in ANT, “translation” can apply to many kinds of practices and objects. Our analysis considered how texts spoke for, stood in for, or seemed to mobilize other texts, objects, or bodies. A paper text in the form of a student assignment, for example, was seen as “[mobilizing] the body and thoughts of the student . . . [continuing] the circulation of the student toward the teacher, where the student could be examined and evaluated” (Leander & Lovvorn, 2006, p. 305).

The heterogeneity of space-time representations, the second dimension, is related to translation. Because texts express particular versions of space-time (time-space or

chronotopes; Bakhtin, 1981; Leander, 2001), we looked for the kinds of space-times particular texts produced and organized linguistically, imagistically, and by other means. In the process of coding and contrasting data, we focused especially on texts that offered participants “representations of their activity that were global in scale with respect to their locally embodied activity” (Leander & Lovvorn, 2006, p. 305) and asked how, by introducing these representations, texts produced multiple time scales (heterochronies; Lemke, 2000) and multiple spaces (heterotopies; Foucault, 1986).

We hoped also to describe a *geography* of texts in activity, through the third dimension, by analyzing how material texts move and fall into order with other actants in activity. By considering the “space-time prepositional relations” (e.g., before, after, during, on top of, besides, behind) of texts to other objects in activity, we could see, for example, that in some forms of activity, the text “is placed” as the final result or curve of what came before (such as writing a summary of a presentation), whereas in other situations texts are “overlaid with activity” (Leander & Lovvorn, 2006, p. 305). Sometimes the space-time relation of texts to other actants may collapse into a text–actant fusion or hybridity. For example, containers that have measurements printed on them, or whose measurements have become routine in a particular practice—such as a half cup in baking—are simultaneously implements and texts of measurement for routine practice (Berg, 1997). According to Berg (1997, p. 420), such “small crossings” are essential for understanding how participants traverse the apparent gap between the formal–textual and practical and, consequently, how material hybridity allows researchers to conceive of texts as acting among a “myriad of little powers.”

Unlike the dimension of movement and position, which foregrounds a spatial dimension of displacement, the analysis of rhythm and speed—the fourth dimension—focuses more directly on temporality. In some kinds of activity (e.g., some computer games), texts and the space-times they mobilize “are placed in frequent and rapid relation to other actants in circulation,” and within such practices, the human participants “appear to beat out a rapid rhythm of texts and action” (Leander & Lovvorn, 2006, p. 306). The speeds are much slower in other kinds of human activity, and the periods between beats may be long.

Finally, with respect to network continuity, texts or other objects in a network might seemingly circulate without end, while other networks are cut off by various means. Where circulations *could* be traced, we gave them “a God’s-eye view” and connected “select, observed locales with myriad other nonobserved locales in space and time” (Leander & Lovvorn, 2006, p. 306). Even from the constrained perspective of a participant, however, some networks are much more continuous than others. By way of comparing Brian’s literacy practices, we considered how his relations to the texts he was reading and writing appeared to be either “more or less extended and continuous” (Leander & Lovvorn, 2006, p. 306), or “cut” (Strathern, 1996).

Selections From Data

The following data from our research is presented in a form that is somewhat “raw,” although, of course, we have already made many analytic moves through processes of recording field notes (and video), selecting these episodes, and summarizing the data. It is also important to reemphasize that for the original article these data were selected from a larger set of ethnographic data, spanning participant observations of several months in Brian’s various classrooms and about a year and a half in his home. We note this because, with the broad look at practices in this research, we attempted in our selection of these episodes to make evident typical social practices and patterns and, thereby, to get at typical ways that literacy was involved in space-time relations in schooling and gaming (our research questions). In this sense, *all* the data (that we are not explicitly writing about) are every bit as important to our thinking as *these* data, because they inform our selections, inform our thinking about the whole, and inform our use of specific constructs from the theory. Although it has been said many times before in relation to ethnography, there is no substitute for sustained engagement over time.

In what follows, we present two summaries of practices from Brian’s classroom and field note summaries from two episodes of video game play at home. These data representations will enable us to illustrate our analytic method subsequently.

1. History Class Lecture and Notes Routine

In his history class, Brian’s teacher, Mrs. Quinn (pseudonyms are used throughout), would present to the class on most of the observed days with a style that combined lecturing with storytelling. When the students arrived in class, their task was to copy questions from the whiteboard pertinent to that day’s lecture (e.g., “How did muckrakers hope to eliminate social evils?”). Students were also to copy vocabulary terms from the whiteboard. To receive credit in class, the overall tasks for the students were to capture the answers to the questions that were prestructured for them and to define the vocabulary terms. Once their notes were completed, students were also to mark their answers to the questions with a highlighting pen to save Mrs. Quinn time in checking over their notes. Lecture notes and defined vocabulary accumulated from day to day, and students were supposed to end up with a large set of notes from which to study for each unit test. Brian enjoyed Mrs. Quinn’s stories during the lecture quite a lot and described history as a “real active class.” However, he was uninterested in taking notes and only did so for class credit. Brian seemed to labor with handwriting while note-taking, often falling behind, and was often missing answers to questions and vocabulary definitions. Brian’s notes were also poorly organized in his notebook, and he reported never using them to study. Nevertheless, Brian was able to earn a B+ in history because, despite his poor performance with note-taking, his test scores were high.

2. English Class Research Project

Brian was uninterested in English class, seeing himself as one who never did well in it. At the same time, Brian never spoke about English class as being challenging. Ms. Marshall (pseudonym), Brian's English teacher, assigned the students a research project in early May that she hoped would help them with their writing. Because of time constraints, the teacher eventually decided that the students would not be able to write the paper and would only complete notecards and an outline. When the class topic was bracketed as "historical periods," Brian chose World War II, because he "just thought there would be a lot of information on it." The students received handouts from Ms. Marshall on how to paraphrase and how to create notecards and made visits to the library and computer lab over a few days. Brian mostly talked with friends in the library and then completed his notecards in his study hall. His notecards offered scanty detail and many of them included information that Brian already knew. One card, for instance, listed "WWII" and then underneath this heading, "Pearl Harbor," followed by "Japan bombed Hawaii." On the day that the cards were due in class, Brian had forgotten about the outline but was able to hastily draft up an outline while Ms. Marshall was circulating around the room and collecting other students' projects. Brian received a B- on the project, scoring full credit for handing it in on time and having completed most of the note cards.

3. (At Home) Video Game Play Episode 1

Observation 2, July 2. Just a week previous to this episode, Brian had created a virtual character for game play, Tiumbe, using a set of tools built into the game. He had also done a bit of exploring around the planet Tatooine. In this session, Brian decided to explore a different planet, Naboo. Brian had already sold the first gun he had purchased, replacing it with a custom-built gun from someone in the game "who [was] really good at building them." Brian had also given Tiumbe new armor on his legs and a new backpack. Tiumbe's exploration of Naboo was motivated by Brian's desire to find a place for his future house, and because Naboo was full of waterways and islands, exploration consisted of a good deal of swimming and walking long distances. Few characters and creatures existed in the areas that Tiumbe explored. Although Tiumbe covered 6,000 meters, as measured on the planetary map, day and night cycled every hour in real time. Brian decided that the islands that Tiumbe explored were too far from the city (and thus other players, shopping, transportation, and other services). Tiumbe headed back to the city to catch a ship to Tatooine while simultaneously communicating with another player whose character was on a different planet in the game. Using a chat screen, he asked how to autorun (make Tiumbe run across the landscape automatically). The other player did not know. Tiumbe then made his way to the starport and, while waiting for his ship, browsed auction listings on a computer terminal located near the star port.

4. (*At Home*) Video Game Play Episode 2

Observation 3, July 10. Brian began by noting the house of a friend, who was a rich player in the game. Tiumbe toured the house and shopped for a sofa among the various items this other player offered for sale. After Tiumbe moved outside, Brian provided a brief explanation of experience points and hunting to the researcher. Tiumbe then teamed up with another player, Ben, to conduct a hunt. The two engaged a giant peko peko bird but were overpowered. To recover, Ben made a camp, but the camp actually disappeared temporarily—an occurrence that Brian explained as a bug in the game. To work around this and other game bugs, Ben decided to log off temporarily. Tiumbe then showed the researcher a new guildhall—a central meeting space for fellow guild members and him. When Ben logged back in, the pair briefly aided other hunters and then successfully hunted a falumpasset (large four-legged creature) by themselves. After the two harvested animal parts, Brian made camp by drawing out a ready-made campsite from Tiumbe’s inventory and noted another game bug: Tiumbe sometimes did not sit directly on the camp chair. Following some surveying work, Tiumbe helped Ben defeat two ikopi (horse-like creatures). After harvesting resources from the animals, the two were beset by rabid shaupauts (kangaroo-like creatures). During this attack, Brian noted, “I’m trying to get experience for trapping,” and so Tiumbe employed traps to help defeat two waves of shaupauts while Brian distinguished between the various types of traps and their effects. Brian then discussed and demonstrated trap crafting, using the resources collected from animals. Tiumbe and Ben then went on additional hunts for other animals, set up a new camp, and crafted additional traps.

Analysis in Practice: Making Sense of the Data and Theory

These data look like quite a mess at first glance, so one impulse might be to tame them with different kinds of models or frames. For instance, for one set of goals, coding themes across the data would be perfectly acceptable. But in this case, we’re trying to keep some of the mess alive, keep looking at materials, and keep looking at movements. We’re keeping our focus on circulations that may give some clues as to the relations between literacy and space-time. To illustrate our process, let’s consider the second dimension of our concept list from ANT: “Heterogeneity of space-time representations.” We believe that not every dimension will work well for all the data, and each dimension may need a lot of refinement for our analysis and for that of others going forward. The eventual description of “Heterogeneity of space-time representations” in our 2006 article arose from our work with the dimension. When we began, what we knew was to look for “heterogeneity of representations.” That concept, then, would come to mean something for us in the context of our work, so one goal became making its meaning

clear to others. (Whether our meanings about it may transfer well to other research, we can't say yet.)

In looking at these data now, with the benefit of hindsight, what we really see are heterogeneous representations (e.g., local and global texts coming together) and heterogeneous mixes (e.g., local actions being combined with more global texts). The simplest case to think with in this dimension is that of the note cards for the English research project. In this case, Brian copies relatively abstract propositions, coming to him from distant sources (e.g., authorities in the library, sources online) and renders that information still more abstract through the processes of making note cards and an outline for an essay. This abstraction and extraction process seem linked to a testing/evaluation economy, as evidenced by Ms. Marshall's concern for the standardized writing assessment and by her use of standardized guides for practices such as note cards. In this regard, Brian's schoolwork in history is quite similar—he extracts notes from the history lectures to answer pre-given questions and predetermined definitions. The knowledge that Brian is tasked with transcribing/re-presenting is given as general, timeless, and spaceless knowledge.

On the other hand—and this contrast lit up as really important to us—Mrs. Quinn, Brian's history teacher, weaves personal stories into her lectures of history. These stories juxtapose abstract representations of the past with local and personal accounts. Brian loves these stories—they are what seem to move him to keep engaged. These local-global connects (Brandt & Clinton, 2002) seem, in ANT terms, to “enroll” Brian as a history student. Although Mrs. Quinn's verbal stories and anecdotes don't continue in the circulations of papers and tests in the classroom, they serve to help Brian continue as an identity pulled into and moved by history. This difference between Brian's English and history courses also illustrates, again, the power of a comparative method for research, in this case not assuming that work practices and circulations look the same merely because they are situated in school. Rather, the two classes, at least in these selected activities, create different relations of space-time to literacy.

The gaming episodes we recorded in our data involved much more heterogeneity of space-time representations than did the classroom data—in fact, we could readily say that this heterogeneity, in terms of local and global perspectives on activity, was the norm. In the episodes summarized earlier, for example, Brian (as Tiumbe) explored Naboo in virtual embodied form, swimming and walking long distances. Within these navigations (yet not recorded in the excerpt above), Brian used waypoints that would connect map-based goals to his local, over-the-shoulder view of his character, and he continually circulated between a map (overhead, more global) perspective and his localized, first-person or over-the-shoulder perspective of his character. Moreover, Brian communicated with other players on other virtual planets while playing, thus giving more global reach to his activity and permitting him to multitask with local and more distal concerns. We further noticed how Brian often circulated between a local form of activity and some broad sense of his goals. For instance, in the middle of the camping

scene depicted, Brian mentions wanting to get experience points for hunting. With the click of a button, he would very often make available to his view a menu that showed his point totals for different types of experience. Thus, he engaged in local/global perspectives that involved more than just navigation—they also involved a sense of how local activity types related to developing identity goals.

Contrasted with making notes in English or history, these ways of reading texts, seeing oneself in relation to the world, and thinking locally and globally during activity, seemed striking to us. They made us wonder, even more than before, how diverse representations of space-time in activity, and especially forms of connection between the local and global, may be related to Brian's senses of himself as an actor in the world. The contrasts in this regard also made clearer how it wasn't just the representations that were of interest but also the ways in which the texts that Brian used were juxtaposed with different types of activity that made such sharp distinctions between gaming and schooling for Brian. For the most part, schooling for Brian involved the circulation of one text to another—such as from the information in the teacher's talk to the notes on his paper or from the library book to his note card. Materially speaking, the work of taking up a text was to produce a different text. In gaming, however, texts were present frequently, but these texts (as images and words) were posed in relation to all types of activity (e.g., navigating, fighting, hunting) and appeared as embedded or dynamically connected to these forms of activity.

Our thinking about the data through the lens of heterogeneous space-time representations (Dimension 2) led us naturally to think about them through the lens of movements and positions of texts in circulation (Dimension 3). One would expect these relations because it's important to keep in mind that these different dimensions don't exist in the world as separate ideas. Rather, they are heuristics or little tools that we have used to tease out some different ways of thinking and seeing from complicated theory. One analytic method that helped us even further consider contrasts related to the movements and positions of texts was to create small sketches for ourselves. Over time, we have come to have a lot of faith in sketching relations as a means to understand data. Sketching can, of course, involve visually organizing data or making provisional models of some concept in relation to the data. In this case, we used sketches at a more basic level—we simply drew lines between texts, other materials, and people in the circulations of Brian's classroom activities and game play. In the backs of our minds was a quote by some educational sociologist (whom we couldn't recall and never quoted in the paper) that the circulation of school was mostly good at producing garbage—vast amounts of student work that ended up in dumpsters. So, we thought, trace that back . . . Where does the garbage come from? Who makes the first text or object? What does it turn into (Translation, Dimension 1) next? Our sketches were powerful tools for helping us understand how the gaming circulations set up incredibly different relations of texts to action and, additionally, created an entire game-based economy related to texts, objects, and activity. Each world Brian

was in, of course, had its own economy, and to understand these circulations, we needed to visualize them.

One issue that we found incredibly intriguing as we contrasted and spent time with our data, reading and rereading, sketching and pondering, was how certain material objects in Brian's game play functioned as what we came to call "text-objects." The animal traps in the second game episode presented earlier are a good example of this phenomenon. Analytically, we want to note that the traps did not attract our attention because we were looking through a theoretical "frame," because we were coding something, or even because we were focused on the question of space-time. The traps were just cool. (Sometimes the data just scream for your attention. They even seem to mock you: "Okay, make sense of ME now, will ya buddy?") On one hand, the traps are representations, both visual and textual. During the middle of an attack by shaupauts, Brian accessed various images of traps in a help screen, clicking on them to pull up more screens that were filled with textual information about the effects of the traps. The traps themselves also provided Brian access to information concerning how many experience points each one would give him—capital toward his identity goals as a player. And, like texts more than objects, the traps could be packed along with all of his gear and seem to have no volume or mass. On the other hand, however, when he needed them, the traps were Brian's faithful tools in the struggle—they functioned like real materials in the scene of the hunt. The ways in which the text-object ontology made a difference in game play really struck us as we considered it, and we worried, "Are we just describing virtual reality? Wasn't all this most obvious?" Perhaps that was the case, but in these parts of the analysis, we have learned over time to trust our instincts in terms of what we find surprising, powerful, or of a striking contrast. The traps, and many other objects in the game, including Brian's own virtual body (as Tiumbe), were not merely texts or merely objects; they functioned and were used as something in between. Their being-in-the-world, or ontology, was different.

The text-object relation was somewhat captured in the third point of our space-time dimensions: "movements and positions of texts in circulation." Within that dimension, we came to describe how there could be a "fusing or hybridity of text-actant" wherein a text and a non-textual actant circulate in strong, rapid fashion (Leander & Lovvorn, 2006, p. 306). From ANT, we could see the text-object as not only *within* a circulation (around the game world) but also, itself, *was* a circulation. We were reminded that one theorist of ANT (Latour, 1991) said to notice how every actant is itself a network, a circulation. The traps, then, seem like a chain of semiotic and material relations (S-M-S-M-S-M) in how they are taken up in the game. On one hand, this seemed like a possible "definition"—a possible holding place for the analysis. Yet, it also felt incomplete—we were (and are) still bothered by the traps, still bothered by how to understand virtual bodies and materials and their relations to meanings and doings. A series of new questions opens up. Is the limit *Actor Network Theory*? Maybe. We are intrigued by how some of the newly-developing work in new materialisms might guide us. Is the limit

the data we have? Perhaps. Now, and especially with artificial intelligence, we have still newer problems to consider around text–objects in our lives, problems that we find both puzzling and exciting.

Moving Forward: Helpful Stances for Literacy Researchers Employing ANT

We'd like to conclude with a short consideration of some stances that we believe are useful to researchers who want to think with ANT. Obviously, we don't believe that thinking *with* ANT is close to being settled in the field, nor do we feel a great deal of confidence that this list is somehow complete. Yet these are qualities that we believe will help you orient to the data in ways that will make more evident the kinds of things ANT is good for—showing human/nonhuman relations, for example, or making clear how different relations of literacy to space-time are created through different forms of circulation. As we have indicated throughout our discussion, we also see these forms of thinking with ANT as contributing potentially to the theory and to the data analysis—they are in some kind of dance of meaning, some kind of circulation. This stance contrasts with some other stances that would ask one to look through the theory as an established lens that is separate from the data.

- **You should value an ethnographic attitude.** In line with Latour's earlier comments about "ethnomethodology," Hamilton (2012) points out that "ANT's preferred method is ethnographic" (p. 43). But as already noted, any ANT ethnography must incorporate an expanded sense of the social to include material elements (Latour, 2005). Researchers accustomed to collecting human-centered data must stretch their gazes and pay greater attention to objects. As far as the nuts and bolts of collecting data, much has been made of Latour's (2005) call to "follow the actors" (p. 68), but ANT researchers often begin with traditional methods of qualitative research, common to ethnography, as ways to identify the actors worth following. Moreover, ANT shares with ethnographic reporting a bias toward storytelling, although ANT's stories often involve narratives of things such as vacuum pumps, laboratory slides, or scallops.
- **You should be attuned to connections and resulting networks.** In "Technology Is Society Made Durable," Latour (1991) writes, "We are never faced with objects or social relations, we are faced with chains that are associations of human (H) and non-humans (NH)" (p. 110). In an ANT account, these *relations*, *chains*, or *associations*, also captured

more broadly with language such as *assemblages* or the eponymous *networks*, are actually what give rise to the work traditionally ascribed to human intention. ANT consistently emphasizes relations and relational sets, and researchers therefore ask questions like the following: *How* are associations among people and things accomplished? (Nespor, 2012), and what do texts and objects *do* in their respective “assemblies or gatherings”? (Fenwick & Edwards, 2012, p. xi). An ANT researcher must function, to some extent, as an on-the-ground detective, tracking and tracing relations as they arise, and seeing where they go.

- **You should become comfortable with complexity.** ANT acknowledges complexity by granting notions of agency and collaboration that are highly varied and often densely intertwined. Speaking to this issue, Mol and Law (2002) write:

Much recent work in the sociology of science, history of technology, anthropology of medicine, cultural studies, feminism, and political philosophy has been a revolt against simplification. The argument has been that the world is complex and that it shouldn't be tamed too much—and certainly not to the point where simplification becomes an impediment to understanding. (p. 1)

Law (2004) continues to press this line in social science research, pointing out how “standard research methods” sometimes produce the reality they seek to understand (p. 4), and he instead advocates for approaches to data that entertain divergent explanations and a wide range of accepted evidence. Indeed, ANT researchers may wind up tracking multiple relations and multiple networks, sometimes with no clear convergence in sight.

- **You should pay attention to texts and to their relations to other objects and to activity.** ANT has an affinity for texts that literacy researchers, in particular, would be well served to notice. In ANT accounts of scientific and technical activity, inscriptional devices and the texts they produce hold special places, in part, because these elements render static entities more mobile and allow traces of their perception to circulate in ways previously impossible. Such “immutable mobiles” impart certain kinds of power—cognitive, rhetorical, political and otherwise—to those who bear them and use them for various kinds of analyses (Latour, 1990). In one instance, Latour (1999b) tracks a scientific expedition to the Amazon, highlighting ways in which a savanna–forest border is sampled, translated into representations via a

range of scientific devices, and eventually moved off-site into laboratories and, later, into journal pages. This “material-semiotic” account, in which the physical world is reduced to symbolic stand-ins, is replete with texts at the same time it showcases ANT’s preoccupation with how inscriptions often gather up elements and transport them (or *translate* them) across space and time (Law, 2008).

- **You should be attuned to spatiotemporal considerations.** Nespor (2012) observes how, in addressing various “hybrid collectives” (networks, assemblages, etc.), ANT focuses on “the movement of their constitutive associations across time and space” (p. 2). Indeed, metaphors of mobilization (Callon, 1986) and circulation (Latour 1999b) are integral to ANT’s origins, and as we noted in our prior study, “to be an actant involves shifting in time and space, which involves the translation of actants as they circulate, are recruited, organized, and hybridized with other actants” (Leander & Lovvorn, 2006, p. 296). The ANT researcher, therefore, must be alert to such movements, seeking to understand how uptake and transport create longer chains of association between people and things.

Suggested Further Reading

ANT has a very large corpus by now, although much less so in educational research. We recommend reading the following texts as a substantial way into this means of thinking with ANT about literacy.

- Fenwick, T., & Edwards, R. (Eds.). (2012). *Researching education through actor-network theory*. Oxford, England: Wiley-Blackwell.
- Latour, B. (2005). *Reassembling the social: An introduction to actor-network-theory*. Oxford, England: Oxford University Press.
- Law, J. (2004). *After method: Mess in social science research*. London, England: Routledge.

References

- Bakhtin, M. M. (1981). Forms of time and of the chronotope in the novel (C. Emerson & M. Holquist, Trans.). In M. Holquist (Ed.), *The dialogic imagination: Four essays by M. M. Bakhtin* (pp. 84–258). Austin: University of Texas Press.
- Barad, K. (2003). Posthuman performativity: Toward an understanding of how matter comes to matter. *Signs: Journal of Women in Culture and Society*, 28, 801–831.
- Barrett, E., & Bolt, B. (Eds.). (2013). *Carnal knowledge: Towards a ‘New Materialism’ through the arts*. London, England: I.B. Tauris.

- Bennett, J. (2010). *Vibrant matter: A political ecology of things*. Durham, NC: Duke University Press.
- Berg, M. (1997). Of forms, containers, and the electronic medical record: Some tools for a sociology of the formal. *Science, Technology, and Human Values*, 22, 403–434.
- Braidotti, R. (2013). *The posthuman*. Cambridge, England: Polity Press.
- Brandt, D., & Clinton, K. (2002). Limits of the local: Expanding perspectives on literacy as social practice. *Journal of Literacy Research*, 34, 337–356.
- Brown, B. (2001). Thing theory. *Critical Inquiry*, 28(1), 1–22.
- Brown, B. (Ed.). (2004). *Things*. Chicago, IL: University of Chicago Press.
- Callon, M. (1986). Some elements of a sociology of translation: Domestication of the scallops and the fishermen of St. Brieuc Bay. In J. Law (Ed.), *Power, action, and belief* (pp. 196–233). London, England: Routledge.
- Cole, M., & Engeström, Y. (1993). A cultural-historical approach to distributed cognition. In G. Salomon (Ed.), *Distributed cognitions: Psychological and educational considerations* (pp. 1–46). Cambridge, England: Cambridge University Press.
- Coole, D., & Frost, S. (Eds.). (2010). *New materialisms: Ontology, agency, and politics*. Durham, NC: Duke University Press.
- Dolphijn, R., & van der Tuin, I. (2012). *New materialism: Interviews & cartographies*. Ann Arbor, MI: Open Humanities Press.
- Engeström, Y. (1999). Activity theory and individual and social transformation. In Y. Engeström, R. Miettinen, & R-L Punamäki-Gitai (Eds.), *Perspectives on activity theory* (pp. 19–38). Cambridge, England: Cambridge University Press.
- Fenwick, T., & Edwards R. (2012) Introduction. In T. Fenwick & R. Edwards, (Eds.), *Researching education through actor-network theory* (pp. ix–xxiii). Oxford, England: Wiley-Blackwell.
- Foucault, M. (1986). Of other spaces. *Diacritics*, 16(1), 22–27.
- Gee, J. P. (1990). *Social linguistics and literacies: Ideology in discourses*. London, England: Falmer Press.
- Hamilton, M. (2012). Unruly practices: What a sociology of translations can offer to educational policy analysis. In T. Fenwick & R. Edwards, (Eds.), *Researching education through actor-network theory* (pp. 40–59). Oxford, England: Wiley-Blackwell.
- Harman, G. (2009). *Prince of networks: Bruno Latour and metaphysics*. Melbourne, Australia: re.press.
- Lamb, J. (2011). *The things things say*. Princeton, NJ: Princeton University Press.
- Latour, B. (1987). *Science in action: How to follow scientists and engineers through society*. Cambridge, MA: Harvard University Press.
- Latour, B. (1988). A relativist account of Einstein's relativity. *Social Studies of Science*, 18(1), 3–44.
- Latour, B. (1990). Drawing things together. In M. Lynch & S. Woolgar (Eds.), *Representation in scientific practice* (pp. 19–68). Cambridge, MA: MIT Press.
- Latour, B. (1991). Technology is society made durable. In J. Law (Ed.), *A sociology of monsters* (pp. 103–131). London, England: Routledge.
- Latour, B. (1993). *We have never been modern*. Cambridge, MA: Harvard University Press.
- Latour, B. (1999a). On recalling ANT. In J. Law & J. Hassard (Eds.), *Actor network theory and after* (pp. 15–25). Oxford, England: Blackwell.
- Latour, B. (1999b). *Pandora's hope: Essays on the reality of science studies*. Cambridge, MA: Harvard University Press.

- Latour, B. (2005). *Reassembling the social: An introduction to actor-network-theory*. Oxford, England: Oxford University Press.
- Latour, B., & Woolgar, S. (1986). *Laboratory life: The construction of scientific facts*. Princeton, NJ: Princeton University Press.
- Law, J. (2004). *After method: Mess in social science research*. London, England: Routledge.
- Law, J. (2008). Actor-network theory and material semiotics. In B. S. Turner (Ed.), *The new Blackwell companion to social theory* (3rd ed., pp. 141–158). Oxford, England: Blackwell.
- Leander, K. M. (2001). “This is our freedom bus going home right now”: Producing and hybridizing space–time contexts in pedagogical discourse. *Journal of Literacy Research*, 33, 637–679.
- Leander, K. M., & Lovvorn, J. F. (2006). Literacy networks: Following the circulation of texts, bodies, and objects in the schooling and online gaming of one youth. *Cognition and Instruction*, 24, 291–340.
- Lemke, J. L. (2000). Across the scales of time: Artifacts, activities, and meanings in ecosocial systems. *Mind, Culture, and Activity*, 7, 273–292.
- Mol, A., & Law, J. (2002). Complexities: An introduction. In J. Law & A. Mol (Eds.), *Complexities: Social studies of knowledge practices* (pp. 1–22). Durham, NC: Duke University Press.
- Nespor, J. (2012). Devices and conceptual change. In T. Fenwick & R. Edwards (Eds.), *Researching education through actor-network theory* (pp. 1–22). Oxford, England: Wiley-Blackwell.
- Spinuzzi, C. (2015). Symmetry as a methodological move. In P. Lynch & N. Rivers (Eds.), *Thinking with Bruno Latour in rhetoric and composition* (pp. 23–39). Carbondale: Southern Illinois University Press.
- Strathern, M. (1996). Cutting the network. *Journal of the Royal Anthropological Institute*, 2, 517–535.
- Street, B. V. (1984). *Literacy in theory and practice*. Cambridge, England: Cambridge University Press.